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ANGULAR BALL BEARING

Abstract

An angular ball bearing includes an outer ring having an outer ring raceway surface, an inner ring having an inner ring raceway surface, and a plurality of balls being rollable between the outer ring raceway surface and the inner ring raceway surface at a contact angle, a bearing ring of at least one of the outer ring and the inner ring being a press-formed product of a steel plate. The bearing ring includes an extending portion extending in an axial direction from the raceway surface toward a side opposite to a side on which a line of action of the contact angle relative to a center of the ball passes through the raceway surface. The extending portion has a bent portion bent in a radial direction and the bent portion overlaps the ball when viewed from the axial direction so as to define an overlap allowance.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to an angular ball bearing, and more particularly to an angular ball bearing in which at least one bearing ring of an outer ring and an inner ring is formed by pressing.

BACKGROUND ART

[0002] An example of an angular ball bearing used for a clutch mechanism or the like of an automobile includes a bearing that can prevent separation of an outer ring or an inner ring after completion of assembly and support integral handling of the bearing. Such a bearing has an advantage of being easy to convey or be assembled to an apparatus.

[0003] Patent Literature 1 discloses a wheel bearing device including an outer member, an inner member (a hub wheel and an inner ring), and double rows of balls interposed between the outer member and the inner member, in which a raceway surface and convex portions provided in a vicinity of the raceway surface are processed by grinding at the same time relative to the inner member made of steel, and the bearing would not be separated due to interference between the balls and the convex portions.

[0004] There is further known an angular ball bearing in which a bearing ring (an outer ring and an inner ring) is manufactured by pressing a metal plate member.

[0005] For example, Patent Literature 2 discloses a wheel bearing in which a rolling surface and counter portions provided in a vicinity of the rolling surface are formed at the same time on a bearing ring by pressing or cold-rolling, the rolling surface and the counter portions are freely subjected to grinding after heat treatment, and the bearing ring would not be separated due to interference between the counter portions and balls.

[0006] Patent Literature 3 discloses a ball bearing in which inner and outer rings formed by pressing a steel plate are provided with engagement portions in a part extending continuously with at least one of a bent portion of the outer ring and a radially extending portion of the inner ring, and the outer ring and the inner ring are engaged by the engagement portions to be not separated.

CITATION LIST

Patent Literature

[0007] Patent Literature 1: JP2001-193745A

[0008] Patent Literature 2: JP2009-052709A

[0009] Patent Literature 3: JP2004-197920A

SUMMARY OF INVENTION

Technical Problem

[0010] In the ball bearing described in Patent Literature 1, the convex portions for making the bearing not separated are formed at the same time as the raceway surface by grinding, and a high grinding cost is an obstacle in reducing the cost of the bearing.

[0011] In the wheel bearing described in Patent Literature 2, groove shoulders are filled with a material during the rolling, and thus a thickness of a part through which a contact angle of the groove shoulder passes may be small.

[0012] In the ball bearing described in Patent Literature 3, the outer ring and the inner ring are

engaged by the engagement portions provided on the outer ring or the inner ring, so that the bearing would not be separated, and the outer ring or the inner ring is required for a shape for forming the engagement portions, which may increase an axial size of the bearing.

[0013] The present invention is made in view of the above-described problems, and an object of the present invention is to provide an angular ball bearing that can be manufactured at a low cost, and has good handleability by preventing separation of a bearing ring and balls.

Solution to Problem

[0014] The above object of the present invention is achieved by following configurations. (1) An angular ball bearing, the angular ball bearing including an outer ring having an outer ring raceway surface on an inner circumferential surface of the outer ring, an inner ring having an inner ring raceway surface on an outer circumferential surface of the inner ring, a plurality of balls being rollable between the outer ring raceway surface and the inner ring raceway surface at a contact angle, and a cage holding the plurality of balls at predetermined intervals in a circumferential direction, a bearing ring of at least one of the outer ring and the inner ring being a press-formed product of a steel plate, in which [0015] the bearing ring includes an extending portion extending in an axial direction from the raceway surface toward a side opposite to a side on which a line of action of the contact angle relative to a center of the ball passes through the raceway surface, and [0016] the extending portion has a bent portion bent in a radial direction, and the bent portion is provided in an entire circumference or at least at two portions in a circumferential direction, overlaps the ball when viewed from the axial direction so as to define an overlap allowance.

Advantageous Effects of Invention

[0017] According to the angular ball bearing of the present invention, it is possible to provide an angular ball bearing that can be manufactured at a low cost, and has good handleability by preventing separation of a bearing ring and balls.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0018] FIG. 1 is a sectional view of a main portion of an angular ball bearing according to a first embodiment of the present invention.

[0019] FIG. 2A is an enlarged view of a main portion illustrating a shape of a bent portion of an outer ring, and FIG. 2B is an enlarged view of a main portion illustrating a shape of a bent portion of an inner ring.

[0020] FIG. 3 is a side view illustrating an outer ring and a plurality of balls according to a modification of the first embodiment.

[0021] FIG. 4A is an enlarged perspective view of a bent portion of the outer ring in FIG. 3, and FIG. 4B is an enlarged view of a part surrounded by a dash-dotted line IV in FIG. 3.

[0022] FIG. 5A is a sectional view illustrating a shape of a bent portion of an outer ring according to another modification of the first embodiment, and FIG. 5B is a sectional view illustrating a shape of a bent portion of an outer ring according to still another modification of the first embodiment.

[0023] FIG. 6 is a sectional view of a main portion of an angular ball bearing according to a second embodiment of the present invention.

[0024] FIG. 7 is a sectional view of a main portion of an angular ball bearing according to a third embodiment of the present invention.

[0025] FIG. 8 is a sectional view of a main portion of an angular ball bearing according to a fourth embodiment of the present invention.

[0026] FIG. 9A is an enlarged view of a main portion of an outer ring in FIG. 8, and FIG. 9B is an enlarged view of a main portion of an inner ring in FIG. 8.

DESCRIPTION OF EMBODIMENTS

[0027] Hereinafter, embodiments of an angular ball bearing according to the present invention will be described in detail with reference to the drawings.

First Embodiment

[0028] As illustrated in FIG. 1, an angular ball bearing **10** is, for example, a clutch release bearing used in a clutch mechanism of an automobile, and includes a substantially circular outer ring **20** having an outer ring raceway surface **21** on an inner circumferential surface thereof, a substantially circular inner ring **30** having an inner ring raceway surface **31** on an outer circumferential surface thereof, a plurality of balls **11** that are rollable between the outer ring raceway surface **21** and the inner ring raceway surface **31** and are in contact with both the raceway surfaces **21** and **31** at a prescribed contact angle θ , and a double-supported cage **12** holding the plurality of balls **11** at prescribed intervals in a circumferential direction.

[0029] The cage **12** may be of a cantilever type such as a crown-shaped cage.

[0030] The outer ring **20** and the inner ring **30** are press-formed products of a steel plate.

Specifically, the outer ring **20** and the inner ring **30** are formed by pressing a plate member of an alloy material containing 0.7 wt % to 0.9 wt % of carbon, 0.3 wt % to 0.9 wt % of manganese, 0.3 wt % to 1.0 wt % of chromium, and 0.01 wt % to 0.15 wt % of silicon.

[0031] As illustrated in FIG. 1 and FIG. 2A, the outer ring **20** includes a substantially cylindrical outer ring extending portion **23** axially extending from the curved outer ring raceway surface **21** toward one side in an axial direction (rightward direction in FIG. 1 and FIGS. 2A and 2B), and a flange portion **25** curved in the axial direction and further extending radially inward on the other side in the axial direction (leftward direction in FIG. 1) relative to the outer ring raceway surface **21**.

[0032] The outer ring **20** further includes a step portion **23a** obtained by pressing to deform an inner circumferential surface of the outer ring extending portion **23** constituting the outer ring **20** to an outer diameter side relative to an inner circumferential surface of a part constituting the outer ring raceway surface **21**. The step portion **23a** has a plate thickness t_1 that is smaller than a plate thickness t_1 of the part constituting the outer ring raceway surface **21**. The outer ring extending portion **23** is provided with, on the inner circumferential surface, a counter portion **23b** having a cylindrical surface shape between the outer ring raceway surface **21** and the step portion **23a**. The counter portion **23b** has the same inner diameter as an inner diameter of a groove bottom portion (maximum inner diameter portion) of the outer ring raceway surface **21**.

[0033] Further, the outer ring extending portion **23** is provided with a bent portion **24** over an entire circumference that is obtained by pressing to bend the outer ring extending portion **23** to an inner diameter side and reducing an inner diameter of a tip portion thereof. The bent portion **24** is provided with, at a tip portion of an inner circumferential surface thereof, a chamfered portion **24a** substantially parallel to an axis X of the angular ball bearing **10**. The step portion **23a** is provided between the outer ring raceway surface **21** and the bent portion **24**.

[0034] Similarly, as illustrated in FIG. 1 and FIG. 2B, the inner ring **30** includes a substantially cylindrical inner ring extending portion **33** axially extending from the curved inner ring raceway surface **31** toward the other side in the axial direction, and a flange portion **35** curved in the axial direction and further extending radially outward on the one side in the axial direction relative to the inner ring raceway surface **31**.

[0035] The inner ring **30** further includes a step portion **33a** obtained by pressing to deform an outer circumferential surface of the inner ring extending portion **33** constituting the inner ring **30** to the inner diameter side relative to an outer circumferential surface of a part constituting the inner ring raceway surface **31**. The step portion **33a** has a plate thickness t_i that is smaller than a plate thickness t_2 of the part constituting the inner ring raceway surface **31**. The inner ring extending portion **33** does not include a counter portion, and an end portion of the step portion **33a** on a radially outer side has the same axial position as a center O of the ball **11**.

[0036] Further, the inner ring extending portion **33** is provided with a bent portion **34** over an entire circumference that is obtained by pressing to bend the inner ring extending portion **33** to the outer diameter side and increasing an outer diameter of a tip portion thereof. The bent portion **34** is provided with, at a tip portion of an outer circumferential surface thereof, a chamfered portion **34a** substantially parallel to the axis X of the angular ball bearing **10**. The step portion **33a** is provided between the inner ring raceway surface **31** and the bent portion **34**.

[0037] Regarding the outer ring **20**, the one side in the axial direction is a side opposite to a side on which a line of action L of the contact angle θ relative to the center O of the ball **11** passes through the outer ring raceway surface **21** in the axial direction. Accordingly, the other side in the axial direction is the side on which the line of action L of the contact angle θ relative to the center O of the ball **11** passes through the outer ring raceway surface **21**. Regarding the inner ring **30**, the other side in the axial direction is a side opposite to a side on which the line of action L of the contact angle θ relative to the center O of the ball **11** passes through the inner ring raceway surface **31** in the axial direction, and the one side in the axial direction is the side on which the line of action L of the contact angle θ relative to the center O of the ball **11** passes through the inner ring raceway surface **31**.

[0038] Accordingly, in the outer ring **20**, the bent portion **24** (chamfered portion **24a**) has an inner diameter D_o that is smaller than an inner diameter D_1 of the outer ring raceway surface **21**. The bent portion **24** overlaps the ball **11** when viewed in the axial direction, and defines an overlap allowance δ_1 of $(D_o - D_1)/2$.

The inner diameter D_1 of the outer ring raceway surface **21** is substantially the same as a diameter of a circumscribed circle of the plurality of balls **11**.

[0039] Accordingly, the bent portion **24** abuts on the balls **11** even when the outer ring **20** moves apart relative to the balls **11** in the axial direction, so that the outer ring **20** and the balls **11** would not be separated.

[0040] Similarly, in the inner ring **30**, the bent portion **34** (chamfered portion **34a**) has an outer diameter D_i that is larger than an outer diameter D_2 of the inner ring raceway surface **31**. The bent portion **34** overlaps the ball **11** when viewed in the axial direction, and defines an overlap allowance δ_2 of $(D_i - D_2)/2$.

The outer diameter D_2 of the inner ring raceway surface **31** is substantially the same as a diameter of an inscribed circle of the plurality of balls **11**.

[0041] Accordingly, the bent portion **34** abuts on the balls **11** even when the inner ring **30** moves apart relative to the balls **11** in the axial direction, so that the inner ring **30** and the balls **11** would not be separated.

[0042] The outer ring **20** and the inner ring **30** are formed into desired shapes illustrated in FIG. 1 as a whole by a series of pressing including a step of bending the bent portions **24** and **34** in the radial direction from a plate member of an alloy material or a steel material.

[0043] Thereafter, necessary portions including the outer ring raceway surface **21** and the inner ring raceway surface **31** are subjected to heat treatment including carburizing and quenching, induction hardening, and the like. Further, the outer ring raceway surface **21** and the inner ring raceway surface **31** are subjected to a finishing process such as superfinishing (SF), deburring, and scaling.

[0044] In assembling the angular ball bearing **10**, first, the plurality of balls **11** are assembled into the cage **12**, and then the balls **11** held by the cage **12** are assembled into the outer ring **20** over the bent portions **24**.

[0045] Further, the inner ring **30** is inserted in a state in which the outer ring **20** is thermally expanded to twice the inner ring overlap allowance δ_2 , and the plurality of balls **11** are assembled into the inner ring **30** over the bent portions **34** of the inner ring **30**. Thereafter, when the outer ring **20** is cooled, the balls **11** would not be separated due to the bent portions **24** and **34**.

[0046] The pressing of bending the bent portions **24** and **34** in the radial direction may be performed separately from the pressing of forming a shape except the bent portions **24** and **34** after

the outer ring **20** and the inner ring **30** are subjected to the heat treatment.

[0047] In any case, the manufacturing cost can be reduced since neither the outer ring raceway surface **21** nor the inner ring raceway surface **31** is subjected to expensive grinding.

[0048] The plate thickness t_1 of the step portion **23a** is 40% to 75% of the plate thickness t_1 of the part where the outer ring raceway surface **21** is formed. This is because, when the plate thickness t_1 of the step portion **23a** is less than 40% of the plate thickness t_1 , the bent portion **24** is easy to deform and it is difficult to hold the balls **11**, and when the plate thickness t_1 exceeds 75% of the plate thickness t_1 , it is difficult to form the bent portion **24** by pressing.

[0049] By providing the step portion **23a** having the plate thickness t_1 on the inner circumferential surface of the outer ring **20** and the bent portion **24** having a small plate thickness t_1 , when the bent portion **24** is pressed, the bent portion **24** can be formed without affecting the shape and accuracy of the outer ring raceway surface **21** by deforming the bent portion **24** from the step portion **23a** separated in the axial direction from the outer ring raceway surface **21**.

[0050] For the same reason as the outer ring extending portion **23**, the plate thickness t_2 of the inner ring extending portion **33** is set to be 40% to 75% of the plate thickness t_2 of the inner ring raceway surface **31**. For the same reason as the step portion **23a** of the outer ring **20**, the step portion **33a** is formed on the outer circumferential surface of the inner ring **30**.

[0051] According to the angular ball bearing **10** in the present embodiment as described above, at least one of the outer ring **20** and the inner ring **30** is formed by pressing, and the outer ring raceway surface **21** and the inner ring raceway surface **31** are formed without expensive grinding. Accordingly, the manufacturing cost can be reduced. Since the outer ring **20** and the inner ring **30** are provided with the bent portions **24** and **34** having the overlap allowances δ_1 and δ_2 , respectively, the bent portions **24** and **34** abut against the balls **11** and prevent separation of the balls **11** and the outer ring **20** or the inner ring **30**, thereby improving handleability of the angular ball bearing **10**.

[0052] In the above embodiment, the bent portions **24** and **34** are formed over the entire circumferences of the outer ring extending portion **23** and the inner ring extending portion **33**.

Alternatively, the bent portions **24** and **34** may be formed at least at two portions in the circumferential direction such that the outer ring **20** or the inner ring **30** would not be separated.

[0053] For example, as illustrated in FIG. 3 and FIGS. 4A and 4B, the outer ring extending portion **23** is provided with the bent portion **24** at two portions in the circumferential direction (in this example, two portions having a phase difference of 180°).

[0054] As illustrated in FIG. 4B, a circumferential length L_1 of the bent portion **24** is larger than a pitch P that is a circumferential interval between adjacent balls **11**. Accordingly, the bent portion **24** normally abuts on any of the balls **11** and holds the balls **11**, so that the outer ring **20** and the balls **11** would not be separated.

[0055] As illustrated in FIG. 5A, the bent portion **24** of the outer ring **20** may be provided with, by pressing at the tip portion of the inner circumferential surface thereof, a tapered portion **24b** having a taper angle of $\alpha=3^\circ$ to 10° relative to the axis X of the angular ball bearing **10** and having an increasing diameter toward the one side in the axial direction. Accordingly, the balls **11** are easy to be assembled to the outer ring **20**, and are difficult to come off after the assembly.

[0056] As illustrated in FIG. 5B, the bent portion **24** of the outer ring **20** may be provided with a curved chamfered portion **24c** at the tip portion of the inner circumferential surface thereof. This can prevent the balls **11** from being damaged when the balls **11** are assembled to the outer ring **20**.

[0057] Although not illustrated, for the same reason as that of the bent portion **24** of the outer ring **20**, the tip portion of the inner circumferential surface of the bent portion **34** of the inner ring **30** may also be a tapered portion having a taper angle of $\alpha=3^\circ$ to 10° and having an increasing diameter toward the one side in the axial direction, or be a curved chamfered portion.

Second Embodiment

[0058] Next, an angular ball bearing **10A** according to a second embodiment of the present

invention will be described with reference to FIG. 6. The angular ball bearing **10A** according to the second embodiment is different from the angular ball bearing **10** according to the first embodiment in shapes of the outer ring extending portion **23** of the outer ring **20** and the inner ring extending portion **33** of the inner ring **30**.

[0059] In the present embodiment, the step portion **23a** is formed on an outer circumferential surface of the outer ring **20** by pressing to deform an outer circumferential surface of the outer ring extending portion **23** to the inner diameter side relative to an outer circumferential surface of a part constituting the outer ring raceway surface **21**. Similarly to the first embodiment, the plate thickness t_1 of the step portion **23a** is set to be 40% to 75% of the plate thickness t_1 of the outer ring raceway surface **21**.

[0060] The outer ring extending portion **23** is provided with the bent portion **24** in the entire circumference or at least at two portions in the circumferential direction, and the bent portion **24** has a reduced inner diameter by pressing at an end portion in the axial direction. The inner diameter D_o of the bent portion **24** is set to be smaller than the inner diameter D_1 of the outer ring raceway surface **21**. That is, the bent portion **24** overlaps the ball **11** when viewed from the axial direction, and defines the overlap allowance δ_1 of $(D_o - D_1)/2$.

[0061] The step portion **33a** is formed on an inner circumferential surface of the inner ring **30** by pressing to deform an inner circumferential surface of the inner ring extending portion **33** to the outer diameter side relative to an inner circumferential surface of a part constituting the inner ring raceway surface **31**. The plate thickness t_i of the step portion **33a** is formed to be 40% to 75% of the plate thickness t_2 of the inner ring raceway surface **31**.

[0062] The inner ring extending portion **33** is provided with the bent portion **34** in the entire circumference or at least at two portions in the circumferential direction, and the bent portion **34** has an increased outer diameter by pressing at an end portion in the axial direction. The outer diameter D_i of the bent portion **34** is set to be larger than the outer diameter D_2 of the inner ring raceway surface **31**. That is, the bent portion **34** overlaps the ball **11** when viewed from the axial direction, and defines an overlap allowance δ_2 of $(D_i - D_2)/2$.

[0063] According to the outer ring **20** and the inner ring **30** in the present embodiment, no step portion is provided on the inner circumferential surfaces of the outer ring extending portion **23** and the bent portion **24** formed continuously on the outer ring raceway surface **21** or the outer circumferential surfaces of the inner ring extending portion **33** and the bent portion **34** formed continuously on the inner ring raceway surface **31**. Accordingly, the balls **11** are less likely to be damaged when being assembled to the outer ring **20** and the inner ring **30**.

[0064] Other configurations and operations are the same as those of the angular ball bearing **10** according to the first embodiment.

Third Embodiment

[0065] Next, an angular ball bearing **10B** according to a third embodiment will be described with reference to FIG. 7.

[0066] Also in the angular ball bearing **10B** according to the present embodiment, shapes of the outer ring extending portion **23** of the outer ring **20** and the inner ring extending portion **33** of the inner ring **30** are different from those of the angular ball bearing **10** according to the first embodiment.

[0067] In the third embodiment, no step portion is provided between the outer ring raceway surface **21** and the bent portion **24** of the outer ring **20** or between the inner ring raceway surface **31** and the bent portion **34** of the inner ring **30**, a part where the outer ring raceway surface **21** is formed has the same thickness as the outer ring extending portion **23**, and a part where the inner ring raceway surface **31** is formed has the same thickness as the inner ring extending portion **33**.

[0068] The outer ring extending portion **23** of the outer ring **20** is tapered from a vicinity of a part of the outer ring raceway surface **21** that has a largest diameter to form the bent portion **24**. The inner ring extending portion **33** of the inner ring **30** is tapered from a vicinity of a part of the inner

ring raceway surface **31** that has a minimum diameter to form the bent portion **34**. Accordingly, the bent portion **24** has the overlap allowance **81** with the ball **11**, and the bent portion **34** has the overlap allowance **82** with the ball **11**.

[0069] The angular ball bearing **10B** according to the present embodiment may be formed by the same method as that of the first embodiment, and may be formed by another method.

[0070] For example, the outer ring **20** and the inner ring **30** are formed by pressing a plate member of an alloy material or a steel material, and the remaining parts of the outer ring **20** and the inner ring **30** are formed into desired shapes except the bending of the outer ring extending portion **23** and the inner ring extending portion **33** in the radial direction.

[0071] Thereafter, except the outer ring extending portion **23** and the inner ring extending portion **33**, necessary portions including the outer ring raceway surface **21** and the inner ring raceway surface **31** are subjected to heat treatment including carburizing and quenching, high-frequency quenching, and the like. Further, the outer ring raceway surface **21** and the inner ring raceway surface **31** are subjected to a finishing process such as superfinishing (SF), deburring, and scaling. The outer ring extending portion **23** and the inner ring extending portion **33** may be softened after the heat treatment.

[0072] Thereafter, after the plurality of balls **11** are assembled into the outer ring **20** and the inner ring **30**, the outer ring extending portion **23** and the inner ring extending portion **33** may be pressed to be bent in the radial direction to form the bent portions **24** and **34** in the outer ring **20** and the inner ring **30**.

[0073] In this case, the outer ring extending portion **23** and the inner ring extending portion **33** including the bent portions **24** and **34** has a hardness that is 15% to 30% of the hardness of the outer ring raceway surface **21** and the inner ring raceway surface **31**, and the bent portions **24** and **34** can be easily formed without affecting shapes and accuracy of the outer ring raceway surface **21** and the inner ring raceway surface **31** even when the part where the outer ring raceway surface **21** is formed has the same plate thickness as the outer ring extending portion **23** and the part where the inner ring raceway surface **31** is formed has the same plate thickness as the inner ring extending portion **33**.

[0074] Other configurations and operations are the same as those of the first embodiment.

Fourth Embodiment

[0075] Next, an angular ball bearing **10C** according to a fourth embodiment of the present invention will be described with reference to FIG. **8** and FIGS. **9A** and **9B**. In the angular ball bearing **10C** according to the fourth embodiment, the bent portion **24** in the second embodiment is applied to the outer ring extending portion **23**, and the bent portion **34** in the first embodiment is applied to the inner ring extending portion **33**.

[0076] Specifically, the outer ring **20** is provided with the outer ring extending portion **23** extending from the outer ring raceway surface **21** to the one side in the axial direction, and the outer ring extending portion **23** is provided with the step portion **23a** on the outer circumferential surface. The outer ring extending portion **23** is provided with the bent portion **24**, which has the overlap allowance **81** with the ball **11**, at two portions different in phase by 180°.

[0077] The outer ring **20** includes a cylindrical outer ring small-diameter portion **26** extending from the outer ring raceway surface **21** to the other side in the axial direction, a radial wall portion **27** extending from an axial end portion of the outer ring small-diameter portion **26** to the inner diameter side, and a folded-back portion **28** extending from a radially inner end portion of the radial wall portion **27** to the one side in the axial direction.

[0078] Accordingly, a part opposite to the outer ring extending portion **23** in the axial direction is formed into a substantially U-shaped cross section. A part of the folded-back portion **28** radially overlaps an inner diameter side of a bent portion **34** of the inner ring **30**, which will be described later, via a radial gap.

[0079] The inner ring **30** is provided with the inner ring extending portion **33** extending from the

inner ring raceway surface **31** to the other side in the axial direction, and the inner ring extending portion **33** is provided with the step portion **33a** on the outer circumferential surface. The inner ring extending portion **33** is formed with the bent portion **34** that is pressed to be radially bent over the entire circumference of the inner ring extending portion **33** and has the overlap allowance $\delta 2$ with the ball **11**.

[0080] The inner ring **30** includes an inner ring large-diameter portion **36** extending from the inner ring raceway surface **31** to the one side in the axial direction, and a radial wall portion **37** extending from an axial end portion of the inner ring large-diameter portion **36** to the outer diameter side. The radial wall portion **37** axially overlaps an end surface on the one side in the axial direction of the bent portion **24** of the outer ring **20**, and an outer diameter side outlet gap C3 that is an axial gap is defined between the radial wall portion **37** and the axial end surface of the bent portion **24**.

[0081] The radial gap defined by the overlapping of the folded-back portion **28** of the outer ring **20** and the bent portion **34** of the inner ring **30** includes an inner diameter side inlet gap C1 and an inner diameter side outlet gap C2. The inner diameter side inlet gap C1 is defined between an outer circumferential surface of an end portion of the folded-back portion **28** on the one side in the axial direction and an inner circumferential surface of the bent portion **34**, and the inner diameter side outlet gap C2 is defined between an outer circumferential surface of the folded-back portion **28** and an inner circumferential surface of an end portion of the bent portion **34** on the other side in the axial direction. The outer circumferential surface of the folded-back portion **28** is formed in a cylindrical shape having a constant outer diameter, and the inner circumferential surface of the bent portion **34** is formed in a tapered shape having an increasing diameter toward the other side in the axial direction, and thus the inner diameter side outlet gap C2 is larger than the inner diameter side inlet gap C1 ($C1 < C2$).

[0082] Further, the outer diameter side outlet gap C3, which is the axial gap between the radial wall portion **37** and the bent portion **24** of the outer ring **20**, is set to be larger than the inner diameter side inlet gap C1 and the inner diameter side outlet gap C2. That is, sizes of the inner diameter side inlet gap C1, the inner diameter side outlet gap C2, and the outer diameter side outlet gap C3 increase in this order ($C1 < C2 < C3$).

[0083] Since the inner diameter side inlet gap C1 is smaller than the inner diameter side outlet gap C2, the amount of lubricating oil supplied from the inner diameter side inlet gap C1 to an internal space of the angular ball bearing **10C** can be controlled to an appropriate amount, and the lubricating oil can smoothly flow into the angular ball bearing **10C** from the inner diameter side outlet gap C2. Since the outer circumferential surface of the bent portion **34** is formed in a tapered shape having an increasing diameter toward the other side in the axial direction, the lubricating oil can be shaken off toward the outer ring **20** from the end portion of the bent portion **34** on the other side in the axial direction by a centrifugal force acting on the inner ring **30** that is a rotary ring.

[0084] Since the outer diameter side outlet gap C3 is larger than the inner diameter side outlet gap C2, the lubricating oil inside the angular ball bearing **10C** is smoothly discharged to the outside of the angular ball bearing **10C** without staying inside, and the stirring resistance of the lubricating oil is reduced. In this way, by controlling the amount and flow of the lubricating oil, the stirring resistance of the lubricating oil can be reduced to reduce the torque.

[0085] In particular, since the bent portion **24** of the outer ring **20** is provided at two or more portions in the circumferential direction, the lubricating oil adhering to the inner circumferential surface of the outer ring **20** can be fed toward the outer diameter side outlet gap C3 through a part of the outer ring extending portion **23** that extends along the axial direction.

[0086] Accordingly, according to the present embodiment, the angular ball bearing **10C** can be manufactured at low cost, and has good handleability by preventing separation of the outer ring **20**, the inner ring **30** and the balls **11**, the amount and flow of the lubricating oil can be controlled, and the stirring resistance of the lubricating oil can be reduced to reduce the torque.

[0087] Other configurations and operations are the same as those of the first embodiment.

[0088] The present invention is not limited to the above-described embodiments and modifications, and modifications, improvements, and the like can be appropriately made.

[0089] In the above embodiments, both the outer ring **20** and the inner ring **30** are formed by pressing. Alternatively, the present invention is not limited thereto, and at least one of the outer ring **20** and the inner ring **30** may be formed by pressing as described in the above embodiments.

[0090] As described above, the following matters are disclosed in the present specification. [0091]

(1) An angular ball bearing, the angular ball bearing including an outer ring having an outer ring raceway surface on an inner circumferential surface of the outer ring, an inner ring having an inner ring raceway surface on an outer circumferential surface of the inner ring, a plurality of balls being rollable between the outer ring raceway surface and the inner ring raceway surface at a contact angle, and a cage holding the plurality of balls at predetermined intervals in a circumferential direction, a bearing ring of at least one of the outer ring and the inner ring being a press-formed product of a steel plate, wherein [0092] the bearing ring includes an extending portion extending in an axial direction from the raceway surface toward a side opposite to a side on which a line of action of the contact angle relative to a center of the ball passes through the raceway surface, and [0093] the extending portion has a bent portion bent in a radial direction, and the bent portion is provided in an entire circumference or at least at two portions in a circumferential direction, overlaps the ball when viewed from the axial direction, and defines an overlap allowance.

[0094] According to this configuration, since the bent portion overlaps the ball when viewed from the axial direction and defines the overlap allowance, and the bearing ring having the bent portion is formed by pressing. Accordingly, the angular ball bearing can be manufactured at low cost, and has improved handleability by preventing separation of the bearing ring and the ball. [0095] (2) The angular ball bearing according to (1), wherein the bent portion has a circumferential length larger than a circumferential pitch of the balls.

[0096] According to this configuration, the bent portion is normally engaged with any of the balls, and thus separation of the bearing ring and the balls is prevented. [0097] (3) The angular ball bearing according to (1) or (2), wherein a tip portion of the bent portion on a circumferential surface on a side on which the overlap allowance is defined is formed into a tapered shape having a taper angle of 3° to 10° .

[0098] According to this configuration, the tip portion of the bent portion is formed into a tapered shape, so that the balls can be easily assembled to the bearing ring and hardly separated. [0099] (4) The angular ball bearing according to (1) or (2), wherein a tip portion of the bent portion on a circumferential surface on a side on which the overlap allowance is defined is chamfered in a curved shape.

[0100] According to this configuration, the balls can be prevented from damage when being assembled to the bearing ring. [0101] (5) The angular ball bearing according to (3) or (4), wherein the extending portion has a step portion between the raceway surface and the bent portion, and the step portion is formed on the extending portion on a circumferential surface on the side on which the overlap allowance is defined.

[0102] According to this configuration, the bent portion can be formed by bending at the step portion, and an influence on the raceway surface can be prevented. [0103] (6) The angular ball bearing according to (3) or (4), wherein the extending portion has a step portion between the raceway surface and the bent portion, and the step portion is formed on the extending portion on a circumferential surface on a side opposite to the side on which the overlap allowance is defined.

[0104] According to this configuration, the bent portion can be formed by bending at the step portion, and an influence on the raceway surface can be prevented. [0105] (7) The angular ball bearing according to (5) or (6), wherein the step portion has a thickness that is 40% to 75% of a thickness of the raceway surface.

[0106] According to this configuration, the bent portion can be more easily formed by pressing without affecting the raceway surface. [0107] (8) The angular ball bearing according to any one of

(1) to (4), wherein a part where the raceway surface is formed has the same thickness as a thickness of the extending portion, and the extending portion has a hardness that is 15% to 30% of a hardness of the part where the raceway surface is formed.

[0108] According to this configuration, the hardness of the extending portion is set to be 15% to 30% of the hardness of the raceway surface, so that the bent portion can be easily formed even when the thickness of the raceway surface is the same as the thickness of the extending portion.

[0109] (9) The angular ball bearing according to any one of (1) to (7), wherein [0110] the outer ring further includes an outer ring extending portion extending in the axial direction from the outer ring raceway surface toward one side in the axial direction which is the side opposite to the side on which the line of action of the contact angle relative to the center of the ball passes through the outer ring raceway surface, an outer ring small-diameter portion extending toward another side in the axial direction relative to the outer ring raceway surface, a radial wall portion extending toward an inner diameter side from an axial end portion of the outer ring small-diameter portion, and a folded-back portion extending toward the one side in the axial direction from a radial inner end portion of the radial wall portion, [0111] the outer ring extending portion has a bent portion radially bent at least at two portions in the circumferential direction, [0112] the inner ring includes an inner ring extending portion extending in the axial direction from the inner ring raceway surface toward the side opposite to the side on which the line of action of the contact angle relative to the center of the ball passes through the inner ring raceway surface, [0113] the inner ring extending portion has a bent portion radially bent over the entire circumference, [0114] the folded-back portion has a radial gap between the folded-back portion and an inner circumferential surface of the inner ring extending portion, and overlaps the inner ring extending portion in the radial direction, and [0115] the radial gap at an inner diameter side inlet defined by an outer circumferential surface of an axial end portion of the folded-back portion and the inner circumferential surface of the inner ring extending portion is smaller than the radial gap at an inner diameter side outlet defined by an inner circumferential surface of an axial end portion of the inner ring extending portion and an outer circumferential surface of the folded-back portion.

[0116] According to this configuration, the bearing ring and the balls cannot be separated, the amount and flow of the lubricating oil can be controlled, and the stirring resistance of the lubricating oil can be reduced to reduce the torque.

[0117] The present application is based on a Japanese patent application (No. 2021-067849) filed on Apr. 13, 2021, contents of which are incorporated herein by reference.

REFERENCE SIGNS LIST

[0118] **10, 10A, 10B, 10C** angular ball bearing [0119] **11** ball [0120] **12** cage [0121] **20** outer ring (bearing ring) [0122] **21** outer ring raceway surface [0123] **23** extending portion, outer ring extending portion [0124] **23a, 33a** step portion [0125] **24, 34** bent portion [0126] **24c** curved chamfered portion [0127] **26** outer ring small-diameter portion [0128] **27** radial wall portion [0129] **28** folded-back portion [0130] **30** inner ring (bearing ring) [0131] **31** inner ring raceway surface [0132] **33** extending portion, inner ring extending portion [0133] **C1** inner diameter side inlet gap (inner diameter side inlet radial gap) [0134] **C2** inner diameter side outlet gap (inner diameter side outlet radial gap) [0135] **C3** outer diameter side outlet gap (outer diameter side outlet axial gap) [0136] **L** line of action of contact angle [0137] **P** circumferential pitch of ball [0138] **t1, t2** plate thickness (thickness of part where raceway surface is formed) [0139] **ti, to** plate thickness (thickness of extending portion) [0140] **δ1, δ2** overlap allowance [0141] **θ** contact angle

Claims

1. An angular ball bearing, the angular ball bearing including an outer ring having an outer ring raceway surface on an inner circumferential surface of the outer ring, an inner ring having an inner ring raceway surface on an outer circumferential surface of the inner ring, a plurality of balls being

rollable between the outer ring raceway surface and the inner ring raceway surface at a contact angle, and a cage holding the plurality of balls at predetermined intervals in a circumferential direction, a bearing ring of at least one of the outer ring and the inner ring being a press-formed product of a steel plate, wherein the bearing ring includes an extending portion extending in an axial direction from the raceway surface toward a side opposite to a side on which a line of action of the contact angle relative to a center of the ball passes through the raceway surface, and the extending portion has a bent portion bent in a radial direction, and the bent portion is provided in an entire circumference or at least at two portions in a circumferential direction, overlaps the ball when viewed from the axial direction so as to define an overlap allowance.

2. The angular ball bearing according to claim 1, wherein the bent portion has a circumferential length larger than a circumferential pitch of the balls.
 3. The angular ball bearing according to claim 1, wherein a tip portion of the bent portion on a circumferential surface on a side on which the overlap allowance is defined is formed into a tapered shape having a taper angle of 3° to 10° .
 4. The angular ball bearing according to claim 1, wherein a tip portion of the bent portion on a circumferential surface on a side on which the overlap allowance is defined is chamfered in a curved shape.
 5. The angular ball bearing according to claim 3, wherein the extending portion has a step portion between the raceway surface and the bent portion, and the step portion is formed on the extending portion on a circumferential surface on the side on which the overlap allowance is defined.
 6. The angular ball bearing according to claim 3, wherein the extending portion has a step portion between the raceway surface and the bent portion, and the step portion is formed on the extending portion on a circumferential surface on a side opposite to the side on which the overlap allowance is defined.
 7. The angular ball bearing according to claim 5, wherein the step portion has a thickness that is 40% to 75% of a thickness of the raceway surface.
 8. The angular ball bearing according to claim 1, wherein a part where the raceway surface is formed has the same thickness as a thickness of the extending portion, and the extending portion has a hardness that is 15% to 30% of a hardness of the part where the raceway surface is formed.
 9. The angular ball bearing according to claim 1, wherein the outer ring further includes an outer ring extending portion extending in the axial direction from the outer ring raceway surface toward one side in the axial direction which is the side opposite to the side on which the line of action of the contact angle relative to the center of the ball passes through the outer ring raceway surface, an outer ring small-diameter portion extending toward another side in the axial direction relative to the outer ring raceway surface, a radial wall portion extending toward an inner diameter side from an axial end portion of the outer ring small-diameter portion, and a folded-back portion extending toward the one side in the axial direction from a radial inner end portion of the radial wall portion, the outer ring extending portion has a bent portion radially bent at least at two portions in a circumferential direction, the inner ring includes an inner ring extending portion extending in the axial direction from the inner ring raceway surface toward the side opposite to the side on which the line of action of the contact angle relative to the center of the ball passes through the inner ring raceway surface, the inner ring extending portion has a bent portion radially bent over an entire circumference, the folded-back portion has a radial gap between the folded-back portion and an inner circumferential surface of the inner ring extending portion, and overlaps the inner ring extending portion in the radial direction, and the radial gap at an inner diameter side inlet defined by an outer circumferential surface of an axial end portion of the folded-back portion and the inner circumferential surface of the inner ring extending portion is smaller than the radial gap at an inner diameter side outlet defined by an inner circumferential surface of an axial end portion of the inner ring extending portion and an outer circumferential surface of the folded-back portion.
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